Introduction to Low Level Design (LLD)

1. LLD.

Low-Level Design (LLD) is the process of deciding how an application will be built internally. It defines the classes, objects, their responsibilities, and how they interact with each other.

DSA vs LLD

- DSA**:** Solves individual problems using algorithms and data structures.
- LLD**:** Decides how different parts of an application are organized and how they work together.

Example**:** In a ride-booking app, Dijkstra's algorithm can find a route, while LLD defines how the Rider, Driver, Ride, and routing components work together.

2. Example: Designing a Ride-Booking App (QuickRide).

DSA First Approach

A developer starts by:

- Representing roads and intersections as a graph.
- Using Dijkstra's algorithm to find the shortest route.
- Using a min-heap to help find nearby drivers.

Problem: This solves individual tasks but does not define the complete application structure, such as payments, notifications, user information, and ride management.

LLD-First Approach

A developer starts by identifying the required components:

- User: Stores common user information.
- Rider: Represents a person requesting a ride.
- Driver: Represents a person offering rides.
- Location: Stores geographical information.
- NotificationService: Sends notifications.
- PaymentGateway: Connects to the payment provider.

Next, the developer defines how these components interact and then uses suitable algorithms wherever needed.

Key point: DSA solves individual problems; LLD organizes these solutions into a complete application.

3. Core Qualities of Good LLD

Scalability

Meaning**:** The application should be able to handle an increase in users and requests.

Example**:** QuickRide should be designed so that increasing ride requests does not require rewriting the entire application.

Maintainability

Meaning**:** Code should be easy to understand, debug, and modify.

Example**:** Keeping payment logic separate from driver-matching logic makes it easier to update one without affecting the other.

Reusability

Meaning**:** A component should be usable in different applications when their requirements are similar.

Example**:** A notification component built for QuickRide could also be used in a food-delivery application.

4. LLD vs HLD

High Level Design (HLD) focuses on the overall system architecture, while LLD focuses on the internal code structure.

LLD	**HLD**
Defines classes and objects.	Defines major system components.
Explains how classes interact.	Explains how services communicate.
Focuses on code structure and responsibilities.	Focuses on architecture and infrastructure.
Example: Designing a Ride class.	Example: Deciding how servers handle ride requests.

HLD also covers database choices, server scaling, deployment, and infrastructure costs.

5. Quick Revision

- DSA**:** Solves computational problems using algorithms and data structures.
- LLD**:** Defines the internal structure of an application and how its components interact.
- HLD**:** Defines the overall system architecture and infrastructure.
- Scalability**:** Ability to handle increasing workload.
- Maintainability**:** Ease of understanding and modifying code.
- Reusability**:** Ability to use components in multiple suitable applications.

Remember: DSA solves individual problems, LLD organizes the code, and HLD defines the overall architecture.